

PRODUCT REQUIREMENTS DOCUMENT

EV Grid Optimization Platform

Forecast EV charging demand • Optimize charging schedules • Support infrastructure planning

Document	Value
Version	1.0
Status	Product definition / MVP-ready
Primary users	Distribution System Operators, EV charging operators, energy planners
Secondary users	Fleet managers, site operators, analysts, city/infrastructure planners
Primary outcome	Lower avoidable charging peaks without violating EV charging requirements
Scope	Forecasting + optimization + monitoring + planning analytics

1. Executive Summary

EV adoption creates a new source of highly variable electricity demand. When many EVs begin charging during the same high-demand period, localized transformers, feeders, substations, and other distribution assets can experience avoidable stress. The proposed EV Grid Optimization Platform is a data-driven decision system that predicts charging demand by time and location, identifies congestion risk, and recommends or automatically generates charging schedules that respect EV-user requirements and grid constraints.

The platform combines historical charging behavior, EV/session information, electricity demand, weather, calendar effects, grid capacity, and charging-infrastructure data. A forecasting layer estimates future load; an optimization layer shifts flexible charging within permitted windows; and a planning layer highlights where additional charging capacity or grid upgrades may be required.

2. Problem Statement

As EV adoption increases, large numbers of vehicles charging at the same time can create sudden peaks in electricity demand and put stress on the power grid. Distribution operators need a way to predict when and where EV charging demand will be high and intelligently manage charging loads.

The goal is to build a data-driven system that forecasts EV charging demand and helps optimize charging schedules and infrastructure planning, reducing grid stress while still meeting EV users' charging needs.

One-line product statement: Predict EV charging demand and intelligently shift/manage charging loads to prevent grid congestion and improve EV infrastructure planning.

3. Goals and Non-Goals

3.1 Goals

- Forecast EV charging demand at useful time and geographic granularities.
- Predict peak-load periods and identify locations/feeders at risk of congestion.

- Generate feasible charging schedules that satisfy EV energy/deadline requirements.
- Reduce avoidable coincident charging peaks and grid constraint violations.
- Preserve user-facing charging requirements such as required energy and departure deadline.
- Provide operators with explainable recommendations, forecasts, alerts, and what-if analysis.
- Use historical outcomes to continuously evaluate forecast and optimization performance.
- Support infrastructure planning by identifying recurring capacity bottlenecks and projected demand growth.

3.2 Non-Goals for MVP

- Real-time autonomous control of physical chargers without an approved integration and safety layer.
- Electricity-market bidding or wholesale market participation.
- Battery degradation optimization beyond a configurable simple constraint.
- Vehicle-to-grid (V2G) dispatch as a core MVP capability.
- Replacing utility protection, SCADA, or operational safety systems.
- Guaranteeing perfect demand forecasts; the system must expose uncertainty instead.

4. Product Principles

- Grid-aware: recommendations must respect modeled electrical capacity constraints.
- User-aware: EV charging needs and deadlines are hard constraints unless explicitly marked flexible.
- Explainable: every major recommendation should expose the reason, expected impact, and affected constraints.
- Forecast with uncertainty: show prediction intervals/confidence rather than a single unexplained number.
- Human-in-the-loop: operators approve actions in MVP before any automated dispatch.
- Fail-safe: missing or stale data must never silently produce unsafe control commands.
- Modular: forecasting, optimization, monitoring, and planning should be separable services.

5. Target Users and Personas

Persona	Needs	Key Actions
Distribution System Operator	See upcoming peaks and constrained assets; mitigate congestion.	Review forecast, inspect risk, approve schedules, monitor results.
EV Charging Operator	Coordinate many charging sessions while preserving customer service.	View sessions, capacity, recommendations, exceptions.
Energy / Infrastructure Planner	Understand future charging growth and network bottlenecks.	Run scenarios, compare sites, estimate upgrade needs.
Fleet Manager	Ensure fleet vehicles are charged before departure at acceptable cost/load.	Set fleet constraints, review schedules, monitor readiness.
Data / ML Analyst	Measure model accuracy and data quality.	Inspect features, drift, errors, model versions.

6. Core User Journeys

Forecast next 24 hours: Operator selects region/time horizon → system loads latest data → model predicts EV load → UI shows forecast, uncertainty, peak windows, and risk.

Mitigate an upcoming peak: System detects predicted constraint → identifies flexible sessions → optimizer creates candidate schedule → calculates before/after load → operator approves.

Handle a new charging session: Session arrives with arrival time, departure deadline, energy required, and charger power → system validates constraints → allocates charging slots → updates forecast/optimization state.

Plan infrastructure: Planner selects area and future EV adoption scenario → system projects charging demand → maps recurring bottlenecks → compares candidate capacity upgrades.

Respond to data failure: Data freshness check fails → system marks affected forecasts stale → optimization is disabled or falls back to safe rules → operator is alerted.

7. Functional Requirements

ID	Capability	Requirement	Priority
FR-01	Data ingestion	Ingest historical and near-real-time charging sessions, charger status, grid load/capacity, weather, calendar, and site metadata.	Must
FR-02	Data validation	Detect missing, duplicated, delayed, impossible, or inconsistent records and assign data-quality flags.	Must
FR-03	Demand forecasting	Forecast aggregate EV charging load by site/feeder/region for configurable horizons.	Must
FR-04	Peak detection	Identify expected peak windows and quantify peak magnitude and timing.	Must
FR-05	Risk scoring	Estimate probability/severity of exceeding configurable grid thresholds.	Must
FR-06	Session modeling	Represent each active/future charging session with arrival, departure, energy, max power, and flexibility constraints.	Must
FR-07	Schedule optimization	Shift flexible charging into feasible time slots while satisfying hard EV constraints.	Must
FR-08	Operator approval	Show recommended schedules and allow approve/reject/modify actions.	Must
FR-09	Impact simulation	Compare baseline and optimized load profiles, peak reduction, constraint violations, and unmet energy.	Must
FR-10	Alerts	Alert on predicted	Must

		congestion, severe forecast uncertainty, data failure, and schedule infeasibility.	
FR-11	What-if analysis	Allow operators/planners to test charger growth, capacity changes, adoption scenarios, and flexibility assumptions.	Should
FR-12	Infrastructure planning	Identify recurring capacity-constrained locations and estimate required additional capacity.	Should
FR-13	Audit trail	Record forecast versions, optimization inputs, recommendations, approvals, overrides, and outcomes.	Must
FR-14	Model monitoring	Track forecast error, drift, data freshness, and optimization performance.	Should

8. Data Requirements

Dataset	Important Fields	Purpose
EV charging sessions	session_id, charger_id, arrival, departure, energy_kWh, requested_energy, max_power_kW, status	Model behavior and optimize flexible sessions.
Charger infrastructure	charger_id, site_id, connector_type, rated_power, availability	Capacity and operational constraints.
Grid topology/capacity	site/feeder/transformer ID, capacity_kW/kVA, current load, limits	Grid constraints and congestion risk.
Historical electricity load	timestamp, asset_id, load_kW/kVA	Separate EV load from background demand and model total demand.
Weather	timestamp, temperature, precipitation, humidity, extreme-weather flags	Capture behavioral/load effects.
Calendar	hour, weekday, weekend, holiday, season	Capture recurring demand patterns.
EV/user metadata	vehicle class, battery capacity, preferred charge window, deadline	Improve demand and scheduling constraints.
Location metadata	latitude/longitude or zone ID, tariff/site type	Spatial aggregation and planning.

MVP can begin with historical or simulated data if live utility/charger APIs are unavailable. Synthetic sessions should preserve realistic arrival, departure, energy, and charger-power distributions.

9. Data Quality Rules

- Timestamps must be normalized to a declared timezone and interval convention.
- A charging session cannot have departure earlier than arrival.

- Requested energy and delivered energy must be non-negative.
- Delivered energy should be physically consistent with charging power and session duration within an explicit tolerance.
- Duplicate session identifiers should be quarantined or deduplicated according to source policy.
- Grid capacity and charger ratings must have units explicitly stored.
- Every forecast input snapshot should have a data freshness timestamp.
- Missing critical constraints must result in an infeasible/unknown state rather than an invented value.

10. Forecasting Requirements

Forecast target: EV charging load, measured in kW or kWh per interval, optionally segmented by site, feeder, transformer, region, charger type, or fleet.

10.1 Forecast horizons

Horizon	Use Case	Suggested Granularity
0–6 hours	Operational congestion mitigation	5–15 minutes
6–24 hours	Day-ahead operational scheduling	15–30 minutes
1–7 days	Planning and staffing	30–60 minutes
1–12 months	Infrastructure planning	Daily/weekly

10.2 Candidate model progression

- Baseline: seasonal naive / historical average by hour and weekday.
- Classical/statistical: regression, ARIMA/ETS where appropriate.
- Machine learning: gradient-boosted trees using lag, calendar, weather, EV-session, and grid features.
- Advanced: temporal deep-learning models only if data volume and validation justify the added complexity.

Model selection must be driven by out-of-time validation and operational usefulness, not model complexity.

11. Optimization Requirements

The optimizer receives predicted demand, active/future charging sessions, charger availability, and grid constraints, then determines charging power by time interval.

11.1 Example decision variable

$x(i,t)$ = charging power assigned to EV/session i during time interval t .

11.2 Hard constraints

- $0 \leq x(i,t) \leq$ maximum available charger/session power.
- Charging can occur only while the session is connected/eligible.
- Required energy must be delivered by the user's departure deadline, subject to physical feasibility.
- Unavailable or faulted chargers cannot be assigned power.
- Grid/asset loading must remain at or below configured operational limits in the optimized schedule.
- Any mandatory charging windows or fleet constraints must be respected.

11.3 Objective

Minimize a weighted combination of grid stress, charging delay/disutility, energy cost (if available), and schedule changes. The weights must be configurable and visible to operators.

Illustrative objective: $\text{minimize } \sum_t [\alpha \cdot \text{PeakStress}(t) + \beta \cdot \text{CongestionPenalty}(t) + \gamma \cdot \text{UserDelay}(t) + \delta \cdot \text{EnergyCost}(t)]$

The exact objective function should be finalized after defining the grid operating policy and user-service requirements.

12. Optimization Modes

Mode	Behavior	Use
Peak shaving	Move flexible charging away from predicted high-load intervals.	Primary MVP mode.
Capacity constrained	Respect feeder/transformer limits while scheduling sessions.	Primary MVP mode.
Cost-aware	Shift charging toward lower-cost periods where tariff data exists.	Phase 2.
Renewables-aware	Prefer periods with high renewable availability where reliable forecasts exist.	Phase 2.
V2G	Allow controlled discharge where technically, contractually, and operationally supported.	Future scope.

13. Infrastructure Planning

- Aggregate forecasted EV demand across sites and network assets.
- Identify assets repeatedly operating near configured thresholds.
- Estimate peak contribution under baseline and managed-charging scenarios.
- Allow scenario inputs such as EV adoption growth, charger additions, fleet electrification, and behavioral changes.
- Compare candidate interventions: managed charging, additional chargers, transformer/feeder capacity, or site redistribution.
- Produce an evidence trail showing which assumptions drive each planning result.

14. User Interface Requirements

Screen	Key Components
Operations Dashboard	Current load, predicted EV load, grid capacity, congestion risk, next peak, active sessions, alerts.
Forecast View	Actual vs forecast, confidence interval, peak markers, selectable asset/site/time horizon.
Optimization View	Baseline schedule, optimized schedule, constraint violations, user impact, expected peak reduction, approval controls.
Session Detail	Arrival/departure, energy requirement, assigned schedule, flexibility, charger state, exception status.
Network/Map View	Sites/assets, utilization, risk status, demand hotspots, drill-down.
Planning Workspace	Scenario controls, projected demand, bottleneck assets, capacity gap, comparison of interventions.
Model Monitoring	MAE/RMSE/MAPE or appropriate metrics, error by segment, data freshness, drift indicators, model version.
Audit Log	Who/what/when for forecasts, schedules, approvals, overrides, and configuration changes.

15. Alerts and Notifications

- Predicted grid congestion within configurable lead time.
- Forecast uncertainty above configured threshold.
- Optimizer cannot satisfy all hard constraints.
- Charger/site unavailable or unexpectedly offline.
- Critical input data is stale or missing.
- Actual load materially deviates from forecast.
- Optimization recommendation is awaiting operator approval.

16. Non-Functional Requirements

Area	Requirement / Target
Availability	Target $\geq 99.5\%$ for MVP dashboard/services during defined operating hours; architecture should support higher availability later.
Forecast latency	Generate routine operational forecasts within a few minutes of receiving required inputs.
Optimization latency	Generate a schedule for a typical operating region within an agreed operational SLA; initial target ≤ 60 seconds for MVP scenarios.
Scalability	Support configurable growth from hundreds to thousands of sessions and multiple sites/assets.
Security	Role-based access, encrypted transport, least-privilege service credentials, secrets outside source code.
Privacy	Minimize personally identifiable information; use pseudonymous session/user IDs where possible.
Auditability	All recommendations and operator overrides must be traceable to input/model/configuration versions.
Observability	Centralized logs, metrics, data freshness checks, model health, optimizer health.

17. Suggested System Architecture

- Data Sources $\rightarrow$ Ingestion/API Layer $\rightarrow$ Validation & Feature Pipeline $\rightarrow$ Data Store/Feature Store
- Historical/streaming data $\rightarrow$ Forecasting Service $\rightarrow$ Forecast Store
- Forecast + active sessions + grid constraints $\rightarrow$ Optimization Engine $\rightarrow$ Candidate Schedules
- Candidate Schedules $\rightarrow$ Rule/Constraint Validation $\rightarrow$ Operator Dashboard $\rightarrow$ Approval
- Approved schedule $\rightarrow$ Charger/CSMS integration (future controlled dispatch) $\rightarrow$ Outcome telemetry
- Telemetry + actual load $\rightarrow$ Monitoring $\rightarrow$ Forecast evaluation + optimization evaluation
- Planning data + forecasts $\rightarrow$ Scenario Engine $\rightarrow$ Infrastructure Planning Dashboard

A practical MVP stack can use Python for data/ML and optimization, a relational/time-series capable database for operational data, REST APIs for service communication, and a web dashboard. Specific technologies should be selected based on team skills and deployment constraints.

18. API-Level Requirements

Endpoint Concept	Purpose	Example Inputs
GET /forecast	Retrieve forecast for an	asset_id, start, end, granularity,

	asset/site/region.	model_version
GET /risk	Retrieve congestion risk.	asset_id, horizon
GET /sessions	Retrieve active/future sessions.	site_id, status, time_window
POST /optimize	Generate candidate schedule.	sessions, forecast, constraints, objective_config
POST /schedule/{id}/approve	Approve recommendation.	operator_id, schedule_id
POST /schedule/{id}/override	Apply operator change.	operator_id, changes, reason
GET /scenario	Run/retrieve planning scenario.	adoption_rate, charger_growth, capacity_change
GET /health	Service/data health.	service/model/data checks

19. KPIs and Success Metrics

Category	Metric	Definition
Forecast	MAE	Mean absolute error of predicted EV load.
Forecast	RMSE	Root mean squared error; emphasizes larger errors.
Forecast	Peak timing error	Difference between predicted and observed peak timing.
Forecast	Peak magnitude error	Difference between predicted and observed peak magnitude.
Grid	Peak reduction	Baseline peak minus optimized peak, measured consistently.
Grid	Constraint violations	Number/duration/magnitude of asset limit violations in modeled scenarios.
User	Deadline satisfaction	Share of sessions meeting required energy by departure deadline.
User	Schedule disruption	Amount of charging shifted/delayed relative to baseline.
Optimization	Infeasible sessions	Sessions that cannot be scheduled under stated constraints.
System	Data freshness	Age of most recent required data by source.
Adoption	Recommendation acceptance	Share of operator recommendations accepted, optionally segmented by reason.

Success thresholds should be calibrated after establishing a baseline on representative data. The product should not declare success solely from forecast accuracy; operational impact and user-service constraints are equally important.

20. MVP Scope

- Historical/simulated EV charging dataset and grid-load dataset.
- Data cleaning and feature pipeline.
- Baseline + one production candidate forecasting model.
- 24-hour rolling forecast by site/feeder/region.
- Peak/congestion risk dashboard.
- Optimization for flexible charging using hard session and grid constraints.
- Before/after load visualization.
- Operator approve/reject workflow.

- Basic alerts and audit logging.
- Evaluation notebook/report comparing unmanaged vs managed charging.

21. Phase 2

- Near-real-time charger/CSMS integration.
- Probabilistic forecasting and uncertainty-aware optimization.
- Tariff/cost-aware optimization.
- Renewable-aware scheduling.
- Advanced spatial/network modeling.
- Infrastructure scenario planning.
- Role-based multi-operator workflows.
- Automated model retraining and drift management.

22. Future Scope

- V2G and bidirectional charging.
- Multi-objective optimization across cost, emissions, grid stress, and user preferences.
- Reinforcement learning for adaptive charging policies, only after strong simulator/evaluation infrastructure exists.
- Digital-twin integration for detailed distribution-network simulation.
- Integration with utility DERMS/ADMS/SCADA under appropriate operational controls.
- Predictive placement optimization for new charging infrastructure.

23. Risks and Mitigations

Risk	Impact	Mitigation
Sparse/inaccurate charging data	Poor forecasts	Data-quality layer, baseline model, uncertainty flags, synthetic augmentation for development.
Grid topology unavailable	Weak spatial optimization	Start with site/asset capacity constraints; add network model in later phase.
User requirements conflict with grid limits	Infeasible schedules	Detect infeasibility explicitly; prioritize hard constraints and expose exceptions.
Model drift	Forecast degradation	Continuous monitoring, scheduled evaluation, retraining triggers.
Operator distrust	Low adoption	Explain recommendations and show measurable before/after impact.
Unsafe automated dispatch	Operational risk	Human approval in MVP; strict validation and fail-safe integration.
Privacy concerns	Compliance/reputational risk	Minimize PII, pseudonymize IDs, enforce access control and retention policies.
Over-optimization	User dissatisfaction	Limit schedule movement and enforce service-level constraints.

24. Assumptions

- Charging sessions have at least approximate arrival/departure and energy requirements.
- The system can obtain or estimate capacity limits for the managed assets.
- Time is discretized into fixed intervals for forecasting and scheduling.
- The initial optimization problem can be modeled without full power-flow simulation.
- Operators define acceptable limits for grid stress and user inconvenience.
- The MVP can operate on historical or simulated data before live integration.

25. Open Questions

1. What exact grid assets are being optimized: charging sites, transformers, feeders, substations, or a combination?
2. What geographic resolution is available in the dataset?
3. What charging-session fields are reliably available?
4. Are grid constraints static ratings or time-varying operational limits?
5. Is electricity price/tariff data available and in scope?
6. Which user requirements are hard constraints versus preferences?
7. What forecast horizon and interval are operationally required?
8. Should optimization target peak reduction, congestion avoidance, cost, or a weighted combination?
9. Will the MVP only recommend schedules, or can it send commands to chargers?
10. What benchmark dataset and evaluation period will define the initial baseline?
11. What privacy/retention requirements apply to user and vehicle/session data?

26. Acceptance Criteria

- Given valid historical/session/grid data, the system produces a forecast for the configured horizon and granularity.
- The dashboard clearly distinguishes actual, forecast, uncertainty, and capacity.
- The system identifies predicted periods where modeled load exceeds or approaches configured limits.
- For a feasible optimization scenario, the generated schedule satisfies all hard session and grid constraints.
- The system reports when a scenario is infeasible instead of silently violating requirements.
- The optimized schedule can be compared against unmanaged/baseline charging.
- The comparison reports peak load, constraint violations, delivered energy, deadline satisfaction, and charging movement.
- An operator can approve, reject, or override a recommendation, with an audit record.
- Stale/missing critical data is visibly flagged and does not silently produce a normal-confidence recommendation.
- Forecast and optimization results can be reproduced using recorded model/configuration/input versions.

27. Example End-to-End Scenario

At 16:00, the platform receives active charging sessions and the latest feeder-load data. The forecasting service predicts that EV demand will rise sharply between 18:00 and 20:00. The feeder's modeled safe capacity would be exceeded if all flexible sessions followed their default charging behavior.

12. Risk engine flags the 18:00–20:00 period.
13. Optimizer identifies sessions with sufficient flexibility before their departure deadlines.
14. Charging is redistributed across earlier/later feasible intervals.
15. The system recomputes the aggregate load profile.

16. Dashboard shows baseline vs optimized peak, grid headroom, number of shifted sessions, and deadline satisfaction.
17. Operator reviews and approves the schedule.
18. Actual outcomes are recorded and later used to evaluate forecast and optimization performance.

28. Recommended Development Milestones

Milestone	Deliverable	Exit Condition
M1 — Data foundation	Schemas, ingestion, cleaning, validation, exploratory analysis.	Representative dataset passes quality checks.
M2 — Forecast baseline	Baseline forecast + evaluation pipeline.	Out-of-time benchmark established.
M3 — Forecast model	Candidate ML/statistical model + API.	Model beats or meaningfully complements baseline on agreed metrics.
M4 — Optimization	Constraint-based scheduling engine.	Feasible test cases satisfy hard constraints.
M5 — Dashboard	Forecast/risk/optimization UI.	Operator can understand and review a complete scenario.
M6 — End-to-end MVP	Integrated pipeline + audit + evaluation.	Complete scenario runs from input data to approved schedule.
M7 — Planning	Scenario and bottleneck analysis.	Planner can compare at least two infrastructure/adoption scenarios.

29. Traceability: Problem → Feature → Metric

Problem	Feature	Primary Metric
Unexpected charging peaks	Demand forecasting	Peak magnitude/timing error
Grid congestion	Risk detection + constrained optimization	Constraint violations / peak reduction
User charging requirements	Session-aware scheduling	Deadline satisfaction
Operator uncertainty	Explainable recommendations + uncertainty	Recommendation acceptance / override rate
Future infrastructure bottlenecks	Scenario planning	Projected capacity gap / recurring utilization

30. Final Product Definition

The EV Grid Optimization Platform is a decision-support and managed-charging system for distribution operators. It turns EV charging and grid data into three connected outputs: (1) a forward-looking view of where and when EV demand will occur, (2) feasible charging schedules that reduce avoidable grid stress while meeting user requirements, and (3) planning insights that reveal where additional infrastructure or alternative managed-charging strategies may be needed. The MVP should prioritize trustworthy data, measurable baselines, constraint correctness, explainability, and operator control before moving toward autonomous real-time dispatch.

End of PRD